06

Resource Management

Structuring the hierarchy, governing access, and keeping usage and spend under control

PCA study series · 9 concept slides · 7 scenario questions

The hierarchy: policy flows down, cost rolls up

1

Organization

Root node tied to a Workspace or Cloud Identity domain. Org-wide IAM and organisation policies live here.

2

Folders

Optional grouping by department, team or environment. Can nest; let you delegate admin to a business unit.

3

Projects

Where APIs are enabled, quotas apply and billing is attached. The trust and isolation boundary for resources.

4

Resources

VMs, buckets, datasets and so on. Each belongs to exactly one project and inherits everything above it.

EXAM LENS Model folders on how the company is governed (departments, environments), not on technology stacks.

How inherited permissions really behave

1 Inheritance is additive

A resource's effective IAM policy is the union of its own policy and every ancestor's. Grants accumulate going down.

2 Children can't take away

A broad grant at a folder cannot be narrowed by an allow policy on a project below it. Grant less higher up, or use IAM deny policies.

3 Moving changes access

Move a project to a different folder and it immediately swaps the inherited policies of the old parent for the new one's.

4 Org policies set guardrails

Organisation policy constraints (allowed regions, no external IPs, no service account keys) restrict what any role can do.

EXAM LENS IAM answers 'who can do what'. Organisation policies answer 'what can be done at all'. Exams mix the two deliberately.

Who holds the keys at the top of the organisation

When an organisation node is created, the whole domain gets Project Creator and Billing Account Creator. Tightening that is usually step one.

Organization Admin	Controls the org's IAM policy and its structure; assign to a small admin group.
Project Creator	Lets users create projects. Remove the domain-wide grant; grant it on specific folders.
Folder Admin	Delegates management of one branch of the tree to a business unit.
Billing Admin	Links projects to billing accounts, sets budgets and receives alerts.

Projects carry three identifiers — only one is yours to change

Identifier	Uniqueness	Mutable?	Used for
Project ID	Globally unique, chosen at creation	No	Commands, APIs, resource names
Project name	Need not be unique	Yes	Human-friendly label in the console
Project number	Globally unique, assigned by Google	No	Internal references, default service accounts

EXAM LENS One project per app per environment is the common pattern: it isolates IAM, quotas and billing cleanly.

Quotas protect you from runaway usage — and from surprises

Types of quota

- Rate quotas: API calls per minute or day; they reset on a schedule
- Allocation quotas: how many networks, VMs or IPs a project may hold
- Regional quotas: such as vCPUs or GPUs per region
- View and request increases in the console under IAM & Admin › Quotas

Why they exist

- Stop a bug or attacker from consuming unbounded resources
- Avoid billing shocks from accidental over-provisioning
- Encourage right-sizing and planned capacity reviews
- You can lower a quota yourself to cap spend on a sandbox

EXAM LENS A quota is a limit, not a reservation. Capacity for GPUs in a zone still needs a reservation.

Labels turn raw billing data into answers

1 What they are

Key–value pairs on resources such as team:payments or env:prod. Up to 64 per resource.

2 Team or cost centre

Charge back spend to the owners who incurred it.

3 Environment and stage

Separate prod, staging and dev costs in one project or across many.

4 Component

Break a bill down by frontend, backend, data pipeline.

5 Owner and state

Find who to ask and which resources are ready for clean-up.

6 Flows into billing export

Labels appear in billing data, so reports can group by them.

EXAM LENS Labels organise and report; they do not grant access or change how traffic flows.

Labels and network tags look similar but do different jobs

Labels

Metadata for people and billing

- Applied to many resource types: VMs, disks, buckets, datasets
- Key–value pairs
- Used for filtering, cost reports and automation scripts
- No effect on networking

Network tags

Targeting for the network

- Applied only to VM instances
- Plain strings such as web-server
- Select which VMs a firewall rule or route applies to
- No effect on billing reports

EXAM LENS 'Apply this firewall rule only to the web VMs' → network tag (or a secure tag / service account), never a label.

A cost-control loop that actually works

1

Budget

Set on a billing account, scoped to projects, labels or services. Use an actual or forecasted amount.

2

Thresholds

Alert at, for example, 50%, 90% and 100% — of actual spend or of forecasted spend.

3

Notify

Email billing admins, and publish to Pub/Sub for programmatic reactions.

4

React

A Cloud Run function on that topic can cap services, scale down or detach billing.

5

Analyse

Export billing to BigQuery and build Looker Studio dashboards grouped by labels.

EXAM LENS Budgets only notify. Anything that stops spending has to be automated through Pub/Sub.

Decision cheat sheet: governance and cost

If the scenario says…	Reach for
Teams manage their own projects, central IT keeps control	Folders per team, with Folder or Project Creator granted at folder level
Restrict where resources can be created	Organisation policy constraint on locations
Narrow a permission inherited from a folder	Change the grant at the folder, or add an IAM deny policy
Know which team spends what across projects	Labels + billing export to BigQuery + Looker Studio
Get warned before the month's bill is too high	Budget with forecasted-spend thresholds
Stop spending automatically	Budget → Pub/Sub → Cloud Run function
Deployment fails with quota exceeded	Request a quota increase, or deploy to another region
Hard-cap a noisy API in a sandbox	Lower that API's quota



7 PCA-style scenarios

Read every requirement word, eliminate options that break one, then pick the most managed answer that satisfies them all. Each question is followed by its answer and the reasoning.

Question 1 of 7

Resource hierarchy

A company has Finance, Marketing and Engineering. Each department must create and manage its own projects, but must not be able to touch another department's projects. A central security team needs read access to everything.

A One project per department, with every user granted Owner on their department's project

B A folder per department with Project Creator granted to that department's group on its folder, and Security Reviewer granted to the security team at the organisation

C Separate organisations for each department linked by VPC peering

D Labels by department on every project, and a firewall rule per label

Answer 1: B

B A folder per department with Project Creator granted to that department's group on its folder, and Security Reviewer granted to the security team at the organisation

Why it fits

Folders mirror the departments, so creation rights are scoped to each branch. A single grant at the organisation gives the security team inherited read access everywhere.

Why the others don't

A — Owner is far too broad and departments need many projects, not one.

C — Splits governance and billing; peering is networking, not access control.

D — Labels do not grant or restrict IAM access.

EXAM LENS Mirror the org chart with folders; grant at the highest level that is still correct.

Question 2 of 7

IAM inheritance

The data-eng group is granted Editor on the Analytics folder. On one project inside that folder, an admin sets the group's role to Viewer to limit changes. What access does the group actually have on that project?

A. Viewer, because the project-level policy is more specific

B. No access, because the two policies conflict

C. Editor, because the effective policy is the union of the inherited and project grants

D. Viewer for 24 hours, then Editor after the policy syncs

Answer 2: C

C Editor, because the effective policy is the union of the inherited and project grants

Why it fits

Allow policies are additive. The Editor grant inherited from the folder still applies, so adding Viewer at the project changes nothing.

Why the others don't

A — Specificity does not override inheritance in allow policies.

B — Conflicting grants are not treated as a deny.

D — There is no time-based sync behaviour like this.

EXAM LENS To reduce inherited access, change the parent grant or use an IAM deny policy.

Question 3 of 7

Billing control

A research group's projects must never exceed $5,000 a month. Finance wants billing disabled automatically if that happens, without anyone watching dashboards.

A. Create a $5,000 budget with a 100% email alert to the billing admin

B. Set quotas on every API to zero after the 25th of each month

C. Create a budget that publishes to Pub/Sub and a Cloud Run function that detaches billing when spend reaches the limit

D. Apply a label cost-cap:5000 to each project

Answer 3: C

C Create a budget that publishes to Pub/Sub and a Cloud Run function that detaches billing when spend reaches the limit

Why it fits

Budgets cannot stop spending by themselves. Publishing notifications to Pub/Sub lets code react — here, by disabling billing on the projects.

Why the others don't

A — Sends an email but spending continues.

B — Arbitrary and disruptive; not linked to actual spend.

D — Labels are metadata and enforce nothing.

EXAM LENS Trigger words: 'automatically stop', 'cap' → budget + Pub/Sub + function.

Question 4 of 7

Cost visibility

Leadership wants a monthly dashboard showing spend per product team. Teams share several projects, and each team's resources are mixed together inside them.

A. Move each team into its own billing account

B. Label resources with a team key, export billing data to BigQuery and build a Looker Studio report grouped by label

C. Download the invoice PDF each month and split it manually

D. Use network tags named after each team

Answer 4: B

B Label resources with a team key, export billing data to BigQuery and build a Looker Studio report grouped by label

Why it fits

Labels attribute individual resources to teams even inside shared projects. The billing export makes that data queryable, and Looker Studio turns it into a dashboard.

Why the others don't

A — Heavy change that still cannot split shared projects.

C — Manual and not granular to resources.

D — Network tags do not appear in billing reports.

EXAM LENS Cost per team / env / component → labels + billing export to BigQuery.

Question 5 of 7

Quotas

A release fails with an error that the regional CPUS quota in europe-west1 has been exceeded. The architecture must stay in that region for data-residency reasons.

A Request a quota increase for CPUs in europe-west1

B Create a new billing account and relink the project

C Deploy the extra VMs to us-central1

D Upgrade to a paid support plan, which removes quotas

Answer 5: A

A Request a quota increase for CPUs in europe-west1

Why it fits

The limit is a per-region allocation quota. Requesting an increase in the same region solves it while respecting residency.

Why the others don't

B — Quotas are tied to the project, not the billing account.

C — Violates the residency requirement.

D — Support plans do not remove quotas.

EXAM LENS Read the constraint: region pinned → raise the quota, don't relocate.

Question 6 of 7

Labels vs tags

Only VMs that serve web traffic should accept inbound TCP 443 from the internet. Other VMs in the same VPC must stay closed.

A Add label role=web to the web VMs and reference the label in the firewall rule

B Add network tag web to the web VMs and target the firewall rule at that tag

C Put the web VMs in a separate project

D Create a budget alert for the web VMs

Answer 6: B

B Add network tag web to the web VMs and target the firewall rule at that tag

Why it fits

Firewall rules select target instances by network tag or service account. A tag on the web VMs applies the rule to them alone.

Why the others don't

A — Firewall rules cannot target labels.

C — Unnecessary restructuring; the VPC rule still needs a target.

D — Billing has nothing to do with traffic.

EXAM LENS Networking selection → tags or service accounts. Reporting → labels.

Question 7 of 7

Guardrails

Compliance requires that no resource in the company is ever created outside the EU, regardless of who has Owner on a project.

A. Remove the Owner role from everyone

B. Apply an organisation policy restricting resource locations to EU regions at the organisation node

C. Set CPU quotas to zero in non-EU regions for every project

D. Write a weekly script that deletes non-EU resources

Answer 7: B

B Apply an organisation policy restricting resource locations to EU regions at the organisation node

Why it fits

Organisation policy constraints apply to everyone beneath the node, including Owners, and block non-compliant creation up front.

Why the others don't

A — Breaks operations and still leaves other roles able to create resources.

C — Covers only some resource types and must be repeated per project.

D — Detects after the fact rather than preventing.

EXAM LENS 'Regardless of role' / 'company-wide' → organisation policy, not IAM.